

ME351 Midterm #1 Take Home    **Feedback Control Systems**    Fall 2025

For this take-home exam, you may use your course notes, your homework solutions, Matlab or Python and your textbook but you may use no other resources. If you consult or collaborate with any of your classmates, all collaborators must include a statement of collaboration with the names of those you consulted/collaborated with and the nature of the collaboration. You may not consult with anyone outside the course. You MUST turn in your own work that reflects your own understanding.  
Try to spend no more than 2 hours doing this take-home exam.

START DAY/TIME: 3pm 1st    END DAY/TIME: 6pm 1st    TOTAL TIME: 3hrs

Include your code, label your plots, and show your work clearly indicating your answers for full credit!

Statement of collaboration with name(s) of collaborators (or include on separate sheet):  
Amber Sun

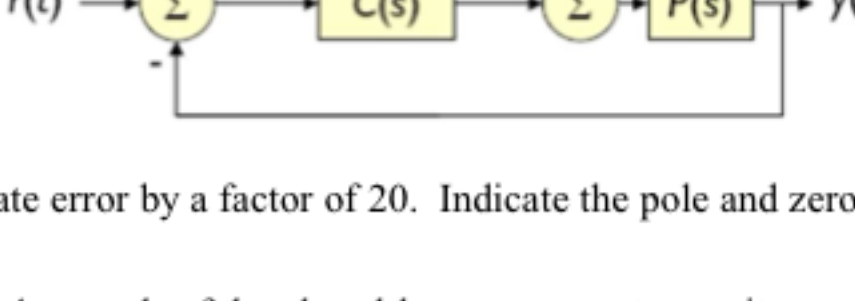
I have neither given nor received any unauthorized aid on this exam, nor have I concealed any violations of the academic integrity code.  
NAME: Dylan Qui    SIGNATURE: [Signature]

1. Reconsider a plant  $P(s) = \frac{1}{s(s+2)}$  and proportional control  $C(s) = K_p$ .

a. If the closed-loop step response has a 10% overshoot, calculate by hand the steady-state error to a **unit ramp input**.

b. Design a lag compensator to improve the steady-state error by a factor of 20. Indicate the pole and zero of the lag compensator.

c. Use Matlab or Python to plot and compare the first 4 seconds of the closed-loop response to a unit ramp reference input for proportional control only ( $K_p$ ) and for your lag compensator. Verify and show the error you calculated in part (b). *Clearly label your plots and turn in your own code.*



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1. CLTF  $\frac{Y}{R} = \frac{CP}{1+CP} = \frac{K_p}{1+\frac{K_p}{s(s+2)}} = \frac{K_p}{s^2+2s+K_p}$      $M_p = 10\% \rightarrow \zeta = 0.6$   
 $K_p = \omega_n^2$      $2\zeta\omega_n = 2$   
 $\omega_n = \frac{2}{2\zeta} = \frac{2}{2 \cdot 0.6} = \frac{2}{1.2} = 1.67$   
 $K_p = \omega_n^2 = (1.67)^2 = 2.78$   
 $E = R - Y = R - bR = R(1-b)$      $R = \frac{1}{s}$  for ramp  
 $\rightarrow \frac{E}{R} = 1-b = \frac{1+2s}{1+CP} - \frac{K_p}{1+CP} = \frac{1}{1+CP}$   
ramp:  $e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \frac{1}{1+CP} \frac{1}{s} = \lim_{s \rightarrow 0} \frac{1}{s^2+2s+K_p} = \frac{1}{K_p}$   
 $e_{ss} = \frac{1}{2.78} = 0.36$   
b. New  $e_{ss} = \frac{e_{ss}}{20} = \frac{0.36}{20} = 0.018$   
lag:  $C(s) = K_p \frac{s+2}{s+p}$      $P(s) = \frac{1}{s(s+2)}$   
 $e_{ss} = \lim_{s \rightarrow 0} s \frac{1}{1+CP} \frac{1}{s} = \lim_{s \rightarrow 0} \frac{1}{1+\frac{K_p(s+2)}{s(s+2)}} \frac{1}{s} = \lim_{s \rightarrow 0} \frac{s(s+2)}{(s+2)(s(s+2)+K_p(s+2))} \frac{1}{s} = \frac{7P}{K_p Z} = 0.01$   
 $\frac{7P}{K_p Z} = 0.01$      $(2 > 1P)$      $\boxed{C(s) = K_p \frac{(s+1)}{(s+0.05)}} \quad K_p = 34$   
 $\frac{P}{Z} = 0.05$     pick  $Z = 0.1$     lag compensator  
 $P = 0.005$

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2. Consider a plant  $P(s)$  under unity feedback where  $C(s)$  is the controller.

Given  $P(s) = \frac{1}{s+4}$  and  $C(s) = \frac{K}{s+a}$

a. Calculate by hand the rise time and overshoot for the case where  $K=40$  and  $a=8$ .

b. For the case where  $a=8$ , determine the gain  $K$  such that the closed-loop system is stable and critically damped.

c. Use Matlab or Python to compare the rise time and overshoot for the  $K$  values in parts (a) and (b).  
*Clearly label your plots and turn in your own code.*

3. A plant has the following transfer function:

$P(s) = \frac{s+2}{s^2+4s+8}$

a. Use Matlab or Python to plot the root locus for the system with unity feedback and proportional gain  $K$  in the forward path. Now change the zero location from -2 to -1, -4, and -20, and compare the root locus for each zero.

b. Use Matlab or Python to find the gain for each with the fastest settling time and compare the steady-state values and overshoot.

c. What might be an undesired real-world consequence of placing the zero at -40 and selecting the gain to achieve the fastest settling time?

a.  $z=2$  all poles & zero on the same vertical line  
 $z=1$  poles to the left of the zero  
 $z=4$  break in/away pts move towards LHP  
 $z=20$  system becomes more stable.

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Given  
1.  $P(s) = \frac{1}{s+4}$      $C(s) = \frac{K}{s+a}$   
2.  $K=40$      $a=8$   
CLTF:  $G = \frac{Y}{R} = \frac{CP}{1+CP} = \frac{\frac{K}{s+a} \frac{1}{s+4}}{1+\frac{K}{s(s+a)}} = \frac{K}{s^2+(a+4)s+4K}$   
 $\rightarrow s^2+4s+8=0$   
 $t_r = \frac{1.8}{\omega_n}$      $\omega_n = 8 \rightarrow \omega_n = 2\sqrt{2}$      $2\zeta\omega_n = 4$   
 $t_r = \frac{1.8}{2\sqrt{2}} = 0.64s$      $\zeta\omega_n = 2$   
 $\zeta = \frac{2}{\omega_n} = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}}$   
 $M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$   
 $M_p = e^{-\frac{\pi \cdot \frac{1}{\sqrt{2}}}{\sqrt{1-\frac{1}{2}}}}$   
 $M_p = 0.04 \rightarrow 4.32\%$   
b.  $s^2+4s-32+K=0$     want  $\zeta=1$  for crit. damped  
 $\sigma > 0$  for stable system  
 $\omega_n^2 = K-32 \rightarrow 4 = K-32$   
 $2\zeta\omega_n = 4 \rightarrow \omega_n = 2$   
 $\omega_n^2 = 4$   
 $K = 36$

RiseTime: 0.7597  
TransientTime: 2.1082  
SettlingTime: 2.1082  
SettlingMin: 4.5172  
SettlingMax: 5.2160  
Overshoot: 4.3210  
Undershoot: 0  
Peak: 5.2160  
PeakTime: 1.5658

RiseTime: 1.6792  
TransientTime: 2.9171  
SettlingTime: 2.9171  
SettlingMin: 0.1175  
SettlingMax: 8.9992  
Overshoot: 0  
Undershoot: 0  
Peak: 8.9992  
PeakTime: 5.9637

3.  $P(s) = \frac{s+2}{s^2+4s+8}$      $C(s) = K$

CLTF:  $\frac{Y}{R} = \frac{CP}{1+CP} = \frac{K(s+2)}{1+K(s^2+4s+8)} = \frac{K(s+2)}{s^2+(4+K)s+8+2K}$

$1+K(s^2+4s+8) = 0 \rightarrow R.L.$

conds: For crit. damping  $\zeta=1$  and  $b^2-4ac=0$   
 $a=1$      $b=4+K$      $c=8+2K$   
 $b^2-4ac=0$   
 $(4+K)^2-4(8+2K)=0$   
 $16+8K+K^2-32-8K=0$   
 $K^2-16=0$   
 $K=4$  or  $K=-16$   
new  $a$      $b$      $c$   
 $\rightarrow$  Quadratic formula to get the best  $K$   
 $E = \frac{1}{1+CP} = \frac{1}{1+K} \frac{s+2}{s^2+(4+K)s+8+2K}$   
 $e_{ss} = \lim_{s \rightarrow 0} s E = \lim_{s \rightarrow 0} s \frac{s+2}{s^2+(4+K)s+8+2K} \frac{1}{s} = \frac{2}{8+2K}$

$\lim_{s \rightarrow 0} Y(s) = \frac{Kz}{b+Kz}$  for step

work for functions in Matlab

|        |           |              |              |                      |              |
|--------|-----------|--------------|--------------|----------------------|--------------|
| z = 2  | K = 4.00  | Ts = 1.348 s | y_ss = 0.500 | Overshoot = 13.531 % | e_ss = 0.500 |
| z = 1  | K = 2.47  | Ts = 2.015 s | y_ss = 0.236 | Overshoot = 52.588 % | e_ss = 0.764 |
| z = 4  | K = 9.66  | Ts = 0.700 s | y_ss = 0.828 | Overshoot = 6.323 %  | e_ss = 0.172 |
| z = 20 | K = 72.22 | Ts = 0.137 s | y_ss = 0.994 | Overshoot = 11.040 % | e_ss = 0.006 |

As  $z$  increases,  $K$ ,  $T_s$  and  $y_{ss}$  all increases. And error steady state decreases. However it is not clear whether there is clear relationship between  $z$  and over shoot % as at  $z = 4$  the overshoot is the smallest out of this set of zeroes.

3C:  
By selecting the gain with the zero so far left, the gain  $K$  is going to be very big at the fastest settling time and that will greatly amplify any noise in the system in real life. This noise being amplified will have undesirable results in real life.